Technician appendix — reference detail for the runbook

This appendix holds the technical reference material that backs each runbook page: setpoint tables, fault-code tables, manual quotations, specifications, service parts and the two procedures that are for a technician only. End users do not need it — every runbook page stands alone. Sections are lettered to match the pointer at the end of each runbook page.

Section	Runbook page
A	Batteries — `cheatsheet-low-battery`
B	Generator — `cheatsheet-generator`
C	Inverter failed — `cheatsheet-inverter-failure`
D	Solar panels — `cheatsheet-array-maintenance`
E	Maintenance calendar — `cheatsheet-maintenance-calendar`

A. Batteries — technician reference

Backs runbook page `cheatsheet-low-battery`.

Sources: Sol-Ark *15K Installation Manual* MA-00007 Rev. 13 (§3.4 Battery Setup, §8.1 Error Codes), the *Pytes V5 User Manual* (§1 specifications, §5.4 start/shut down, §7 troubleshooting), the *Pytes Sol-Ark Guide for V5*, and the installer's customer-training deck. For an LFP bank (Pytes V5) on parallel Sol-Ark 15K-2P inverters.

The one rule that voids the warranty

> *"The battery should be charged within 12 hours when it's fully discharged or over-discharging protection mode is activated. Fail to follow this instruction will damage the battery and is not covered by warranty."*
>
> — Pytes V5 manual, §Safety

Two safeties, in the order they trip

Layer	What it does	Where set
1. Sol-Ark `Shutdown` V/%	Inverter stops AC output *"to protect the battery from an over discharge situation"*; battery icon turns red. Output resumes at **`Restart` V/%**. If grid or generator is available the inverter passes it through instead.	Battery Setup → Discharge
2. BMS under-voltage protection	The pack itself stops discharging (ALM **steady red**, other LEDs off). Pytes V5 enters this **below ~5 %** (installer) but keeps a reserve so it can be charged back. Per the manual, protection **releases automatically** once charge is applied — but only if that happens **within 12 hours**.	Fixed in the BMS; pack range 47.5–57.6 V

Layer 1 is meant to fire first and leave headroom above layer 2. Between them sits **`Low Batt`** (icon turns yellow): on an off-grid site TOU may discharge down to *Shutdown*, on a grid-tied site only down to *Low Batt* (§4). The goal is never to reach layer 2.

Self-clearing alarms (overload, under-voltage, low battery) auto-restart **up to 5 times**; after the fifth stop on the same alarm the inverter **locks out until manually reset** and the cause is fixed (installer deck). **F56** DC_VoltLow_Fault: *"Batteries are overly discharged, the inverter is Off-Grid and exceeded the programmed batt discharge current by 20 %, or Lithium BMS has shut down."* F56 together with **F58** (BMS comms lost) means the BMS has likely opened its contactor.

The Pytes manual is explicit that a blinking or even steady ALM *"does not necessarily indicate a faulty battery"*, and that protection *"will resume normal operation automatically once the 'protection' status is released."* On a pack whose LEDs do not respond to POWER + SW: *"charge the battery correctly... if the battery enters into charging mode, it should return to its normal state after completing the charging process." "Do not repair the battery if no authorization from Pytes!"*

Recommended usable window (training deck)

- LFP: use down to **10–20 % SOC** in normal operation (80–90 % usable). The battery warranty excludes **routine discharge below 10 %**.
- Installer's off-grid setpoints: generator auto-start **35 %**, Sol-Ark Shutdown **15 %** (Sol-Ark's own default is 20 %), emergency floor **11 %**.
- During an outage the bank may go lower before Shutdown — acceptable occasionally, not routinely.
- Bring the bank to **100 % SOC at least once a week** for BMS balancing. Off-grid, that means a clear day with solar — the generator stops at ~95 % and will not do it.

- After any event that let the bank reach BMS protection, review whether the Sol-Ark `Shutdown` setpoint should be raised.

Setpoints to verify (Battery Setup → Discharge)

Setting	Purpose	Sanity check
`Shutdown` %	Inverter stops output	Comfortably above the BMS protection point (~5 % on Pytes) — 15 % is the installer's value
`Restart` %	Output resumes	5–10 points above Shutdown so it does not cycle
`Low Batt` %	Yellow warning; TOU floor on grid-tied	Between Shutdown and your normal daily floor
`Start %` (Gen Charge)	Generator auto-start	Above Low Batt, well above Shutdown — 35 % leaves ~20 points of reserve
Max Gen Runtime	Ends a run by time	Long enough to reach ~90 % from Start % at the set Amps
`BMS_Err_Stop`	Stop on loss of BMS comms	Know what it is set to — it decides what slaves do if the master (and its CAN cable) fails
Capacity (Ah/kWh)	Used for % SOC and limits	Matches the installed bank
Charge Efficiency / Batt Empty V	SOC calculation	Leave at manufacturer values; do not tune
`Use Batt % Charged`, `BMS Lithium Batt = 00`, ☑ `Activate Battery`	Closed-loop Pytes comms	All three required by Pytes's Sol-Ark guide; Sol-Ark manual: Activate Battery *"MUST be selected … especially Lithium batteries."* A stale or missing battery display on the Sol-Ark means one is off or the CAN cable is out
Charge / discharge A (inverter side)	Per-pack limit × packs	Pytes V5: **75 A recommended, 100 A max** continuous per pack; derated outside 10–40 °C cell temperature; **no charging below 0 °C**

Pytes V5 pack facts

51.2 V nominal, 100 Ah, **5.12 kWh** per pack; operating range 47.5–57.6 V. Charge 0–45 °C, discharge −10–50 °C. Cycle life ≥6000 at 90 % DOD; calendar life ≥10 years. Self-discharge 1–2 %/month. Master pack carries the DIP-switch address and the CAN cable to the Sol-Ark (standard Ethernet cable; Sol-Ark *Battery CANBus* jack).

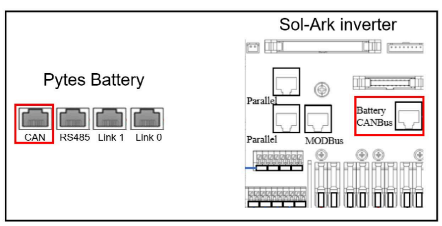


Figure 2.1.1.3 Sol-Ark inverter comm cable connection

Figure 2.1.1.4 Sol-Ark inverter DIP Switch Setting

Pytes Sol-Ark guide, Figures 2.1.1.3 and 2.1.1.4.

Bank power-up order (manual §5.4) — batteries before inverter, always, to protect the packs from the inverters' capacitor inrush: DC breakers on (if fitted) → all POWER buttons ON → press the **master** pack's SW for 1 s → wait for every pack to show lights → only then power the inverters (slaves first, master last). **Shut down:** hold the master SW **3 s** → wait for all pack lights to go out → POWER buttons off → breakers off.

Force charge (last resort): isolate the pack (POWER off, breaker off if fitted) and charge it alone at the terminals with a **51.2 V LFP force charger** until it wakes and shows RUN steady, then return it to the bank. Never parallel a flat pack into live packs. The installer's advice: own that charger before it is needed.

Storage / long idle: keep packs at 40–60 % SOC; if idle more than 6 months, charge to

> 90 % every 6 months. A pack found at zero after storage:
> *"do not charge or use it without permission, contact your installer."*

Set a SolarAssistant or Home Assistant alert a few points above `Low Batt` so the first warning is a notification, not a yellow icon nobody is looking at.

B. Generator — technician reference

Backs runbook page `cheatsheet-generator`.

Sources: Sol-Ark *15K Installation Manual* MA-00007 Rev. 13 (§2.5 Integrating a Generator, §3.4 Battery Setup, §4 Operation Notes), the installer's customer-training deck (as-delivered version), and the Wildcat Patriot / Hyundai *Operation and Maintenance Manual* (20–100 kW mobile generating sets). Off-grid site: prime generator on the **GRID input** with two-wire auto-start from the master's pins 7 & 8; separate 50 A inlet for a portable generator on the **GEN input**.

How it behaves when left alone

Rule	Source
Auto-start fires when the bank reaches **Start V or Start %** (one condition, not both)	§3.4 Gen Charge
Charging from the generator **stops at ~95 % SOC** — *"the batteries will charge until the battery bank accepts 5 % of its rated capacity in Amperes"*. It will **never** reach 100 % on generator power.	§4 note 6
The 95 % ceiling is *"non-modifiable unless Time of Use is enabled and programmed"*	§4 note 6
If TOU is on, the generator will not auto-start in any interval that does not have ☑ **Charge** ticked, *"even if the Start V or Start % condition has been met"*	§2.5
Gen/Grid "A" is per inverter. Multiply by the number of inverters for the current into the bank.	§2.5
Off-grid: keep the **Gen A and Grid A values equal** *"to avoid logic issues"*	training deck
Max Gen Runtime (firmware 7228+, bottom of the Charge tab) and Gen Down Time can end or block a run regardless of SOC	training deck; §3.4
Charge rate **tapers above 90 % SOC** (roughly 10 A per pack between 90 and 95 %). Size Max Gen Runtime to reach ~90 %, not 95 %.	training deck
Installer's off-grid setpoints: auto-start **35 %**, Sol-Ark shutdown **15 %**, Max Gen Runtime **150 min**.	training deck
Weekly exercise: **Monday 08:00, 20 min** by default. Disable with `00 | 00 min`. Runs only if the DSE is in Auto.	§4 note 6; §3.4
With a generator on the GRID input, ☑ **GEN connect to Grid input** must be set, Grid Mode *General Standard*, Grid Reconnect Time 30 s	§3.4 Grid Charge; §4 note 5
GEN terminal continuous limit **80 A** — do not exceed	§2.5
Gen Force is a test function: *"The generator will not provide power during this test if grid power is available"*. Off-grid it runs **until unticked**; it does not stop at 95 %.	§3.4

Adjusting Start % on the fly

Start % is just a setting. On a clear-sky morning with the bank just above Start %, lowering it a few points avoids a pointless run; raise it back afterwards. **Never routinely below ~10 %** (LFP warranty). The installer's stated emergency floor is **11 %**, and only if the generator will not start and the capacity is needed.

Fault matrix

Symptom	Cause	Fix
Generator bogs / breaker trips shortly after qualifying	Charge current + loads exceed generator rating	Lower **Gen A** (per inverter). Shed heavy loads while charging.
Sol-Ark shows AC briefly then rejects it; **F60** Gen_Volt_or_Fre_Fault	Frequency or voltage outside limits — usually overload pulling it under 60 Hz	Lower Gen A; check governor; keep loads light until SOC recovers
Gen Signal on but nothing cranks	Two-wire loop open, DSE not in Auto, Start Failure latched, E-stop in	DSE to Auto; loop is on the **master's** pins 7 & 8 (N.O. dry contact); reset DSE
Gen Signal never comes on	TOU interval without ☑ Charge; Gen Down Time still counting; Max Gen Runtime hit	Check TOU intervals; check Gen Down Time; clear with Gen Force
Charging stops at ~95 % with generator running	Normal cutoff	Set an upper limit via TOU if it must stop sooner
Generator ran for hours, SOC barely moved	Gen A too low for the bank, or heavy loads	Raise Gen A within generator capacity; check load

Rule of thumb for Gen A: generator continuous kW × 1000 ÷ ~55 V ÷ number of inverters, then back off 10–20 % for loads and governor headroom. Portable: ~50 % of generator capacity — e.g. 12 kW → ~6 kW → ~120 A DC total → **60 A per inverter** on two inverters. ~20 A × ~50 V ≈ 1 kW per inverter.

Generator — Wildcat Patriot 40 kW, DSE 6110 MKIII

Ratings (single-phase 120/240 V): 36 kW prime, **150 A** per leg, 50 kVA nameplate. Derate 1.5 % per 10 °F above 77 °F. Fuel: **3.6 gal/h full load, 2.8 at 75 %, 2.0 at 50 %.** Control panel inside the enclosure behind the **left rear door**: control-supply switch, DSE controller, MCCB, emergency stop. (The spec page says "oil change 500 h"; Table 37 puts oil and filter in the 250 h / 1 year column — follow Table 37.)

Auto requirement: Sol-Ark manual — *"the gen must be in automatic mode"*. In Auto the DSE waits for the remote-start contact and runs its own start, warm-up, cool-down and stop timers.

Want to...	On the DSE panel
Start by hand	**Manual** (LED lights) → **Start** once. Run unloaded ~3 min before closing the MCCB.
Stop by hand	Open the MCCB, idle ~5 min, **Stop** once — cool-down timer, then stops.
Clear a fault	**Long-press Stop/Reset** until the fault clears. Fix the cause first.
Return to auto-start	**Auto** key; confirm LED.
Switch the controller off	Open the breakers inside the controller panel.

Start Failure latches after several crank attempts; common causes: low fuel, gelled filter, flat starting battery. **Latching safety stops:** low oil pressure, high coolant temperature, over/under-speed, E-stop, alternator under/over-voltage or over-frequency — the Sol-Ark cannot clear them.

Starting battery: kept up by (a) the generator's own automatic charger, fed with the **magnetic sump heater** (startable below 32 °F; 1000–1500 W) from the **120 V NEMA 5-15 inlet** on the power-panel door, and/or (b) the external **Battery Tender Plus 12 V / 1.25 A** (Deltran 022-0185G-DL-WH), a 4-stage float maintainer, ~20 W, LED (colour-only legend printed on the unit): red steady = charging, green flashing = >80 %, green steady = float, red flashing = fault / reversed clips. For readers who cannot distinguish red from green: steady = floated, flashing = not yet floated, off = fault. Two float chargers on one battery is harmless, but the tender does not replace the inlet in winter — the heater needs it. Keep that circuit on a load that stays up at low SOC. **Disconnect the tender before the battery negative** when working on it.

Disabling for work: battery negative off so it cannot remote-start; untick Gen Force.

Service intervals (Table 37)

You will need — annual service (250 h / 1 year column):

- Engine oil: **SAE 15W-40, API CI-4 or higher**, from sealed containers. Capacity for the 40 kW set (HDI DM03PG engine): **2.5 US gal (9.7 L)** per the specification table; the engine data page states 13.3 qt (12.6 L) — buy **3.5 gal** and fill to the dipstick mark.
- Lube-oil filter (spin-on) — part: see Patriot manual parts list / dealer.
- Primary fuel filter with water separator (1.0 L single-bowl spin-on) — part: see Patriot manual parts list / dealer.
- Secondary fuel filter — part: see Patriot manual parts list / dealer.
- Drain pan (≥4 gal), oil-filter wrench, socket set, torque wrench, funnel, rags, nitrile gloves, sealed container for waste oil.
- Coolant for top-up: **long-life (LLC/ELC) ethylene-glycol, 50/50 with demineralised water** — never tap water, never mix with conventional coolant (the older type usually dyed green).
- Anti-gel additive; fresh **ultra-low-sulfur diesel** only.

You will need — two-year service (500 h / 2 years column), in addition: air-cleaner element, fan/alternator V-belt, radiator and heater hoses and clamps as found worn, feeler gauges for valve clearance — parts: see Patriot manual parts list / dealer. Valve clearance, water pump and thermostat work is for a diesel mechanic.

You will need — walk-around: flashlight, rags, ~1 qt container for the separator drain.

When	Items
Daily / before each run	Walk-around; air-cleaner restriction indicator; oil level; coolant level; drain water separator; hoses and clamps; radiator fins
First 50 h (new engine)	Fuel primary + secondary filters; belt check/adjust; **engine oil + oil filter**
Every 250 h or 1 year	Fuel primary + secondary filters; belt check/adjust; **engine oil + oil filter**; centrifugal filter clean; CCV pipe drain (4K1080TA1 engine only)
Every 500 h or 2 years	Air-cleaner element; belt replacement; hoses/clamps replace as needed; valve clearance; fasteners; engine mounts; starter; charging alternator; water pump; thermostat
Every 1500 h	*(no items listed for these engines)*

Table 37: General maintenance schedule

Service Item	Activity	Frequency	Engine running hours			
		Daily	1st 50 Hrs	Every 250 Hrs / 1 Year (whichever occurs earlier)	Every 500 Hrs / 2 Years whichever occurs earlier)	Every 1500 Hrs
Walk-Around Inspection	-	✓				
Engine Air Cleaner Restriction Indicator	Inspect	✓				
Engine Oil Level	Check	✓				
Coolant Level	Check/Top up if necessary	✓				
Fuel System Water Separator	Drain	✓				
Hoses and Clamps	Inspect	✓				
Fuel System Primary Filter	Replace		✓	✓		
Fuel System Secondary Filter	Replace		✓	✓		
Charging Alternator and Fan Belt	Check/Adjust		✓	✓		
Engine Lubricating Oil	Change		✓	✓		
Centrifugal Filter	Clean			✓		
Lubricating Oil Filter	Change		✓	✓		
Radiator Fins	Check/Clean if necessary	✓				
Engine Air Cleaner Element	Replace				✓	
Charging Alternator and Fan Belt	Replace				✓	

Table 37: General maintenance schedule (Continued)

Service Item	Activity	Frequency	Engine running hours			
		Daily	1st 50 Hrs	Every 250 Hrs / 1 Year (whichever occurs earlier)	Every 500 Hrs / 2 Years whichever occurs earlier)	Every 1500 Hrs
Hoses and Clamps	Check/Adjust/Replace if necessary				✓	
Engine Valve Clearance	Check/Adjust				✓	
Fasteners	Inspect				✓	
Engine Mounts	Inspect/Replace if necessary				✓	
Starting Motor	Inspect				✓	
Charging Alternator	Inspect				✓	
Water Pump	Check/Replace if necessary				✓	
Thermostat Element	Check/Replace if necessary				✓	
CCV Pipe Oil Drain (only for 4K1080TA1 engine)	Inspect/drain if necessary		✓	✓		

Patriot O&M manual, Table 37 (pp. 156–157).

The Patriot: 20kW/40kW/70kW/100kW mobile generating sets

Table 22: Distribution panel

on	Make, part number	Function / instructions
cket 5, /AC	Hubbel, Catalog Number HBL.5278C	• Terminates 120 VAC electrical power from field. • This power energizes the battery charger and sump heater.
t 1 AC,	Hubbel, Catalog Number CS6369	• The plug has a locking mechanism. • Insert the corresponding plug into the receptacle and twist to lock it in place.
t 2 AC,		
0 3 le	Schneider, A9N2P50C	Overload protection and isolation switch for XC1 receptacle.
0 3 le		Overload protection and isolation switch for XC2 receptacle.
t 3 20	Hubbel, Catalog Number CS6369	• The plug has a locking mechanism. • Insert the corresponding plug into the receptacle and twist to lock it in place.
t 4 20		
20 3 C60	Schneider, A9N2P20C	Overload protection and isolation switch for XC3 receptacle.
0 3 C60		Overload protection and isolation switch for XC5 receptacle.
nal m. DC,	Sonax	• Stud terminal for high-current power connections, especially in environments with severe vibration. • Terminate on wires with crimped ring or fork lugs. Secure the lug on the threaded stud with a hex nut torqued to the required value.

Patriot O&M manual, Table 22. Item 1 (X9) is the inlet the sump heater and battery charger depend on.

Calendar limits apply ("whichever occurs earlier"): the 250 h items are an **annual** service and the 500 h items a **two-yearly** one regardless of hours. Log in Table 38. After start: watch oil pressure, coolant temperature, battery voltage, frequency (60 Hz) and load within rating. **Cold weather:** anti-gel in every fill unless the tank will be empty before freezing; gelled-fuel pump damage is not warranted; drain the separator more often; keep the sump heater powered.

C. Inverter failed — technician reference

Backs runbook page `cheatsheet-inverter-failure`.

Source: Sol-Ark *15K Installation Manual* MA-00007 **Rev. 13** (July 2026), §2.12 Power Cycle Sequence, §5 Parallel Systems, §8.1 Error Codes. Applies to 120/240 V split-phase parallel systems on a shared battery bank. Site: 2 × 15K-2P, 12 × Pytes V5, off-grid, Patriot 40 kW on the GRID inputs.

What the master does that a slave doesn't

Responsibility	Manual reference
Holds **Modbus SN 1**; its GRID and Battery settings are copied to every slave	§5.2 step 3; §4 power-on step C
Controls the **generator two-wire start** (relay on pins 7 & 8)	§5.2 note: *"The inverter assigned as Master will control the two-wire start feature"*

Responsibility	Manual reference
Carries the **CT pair** (grid-tied sites only)	§2.9: *"Only one pair of CT sensors must be wired to the designated Master inverter"*
Is the only unit whose **Battery CANBus** jack is cabled to the BMS on a typical build — slaves take battery state over the parallel link	Site wiring convention; see page 00

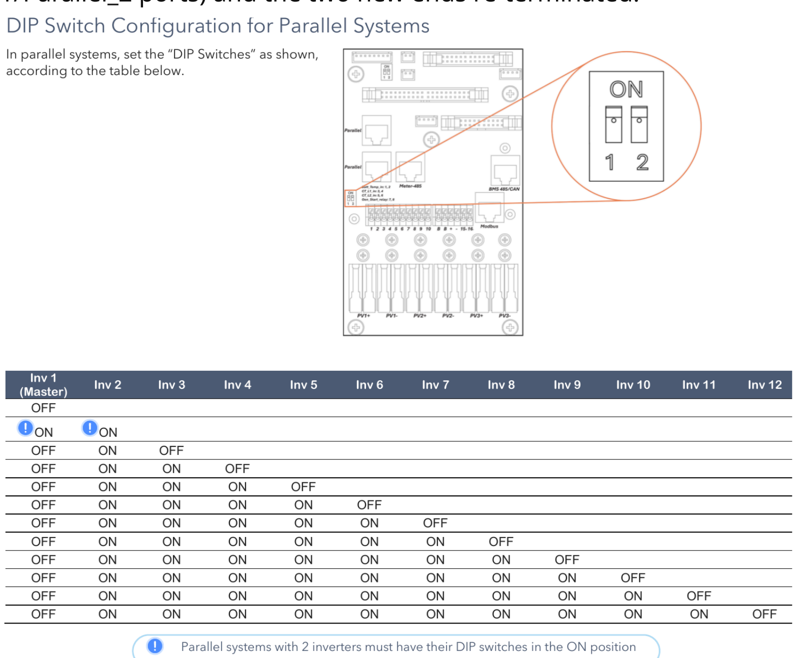
Everything else — LOAD, GRID, GEN AC connections, the battery bank — is shared by every unit (§5.1 C: *"All INPUTS/OUTPUTS must be shared among ALL parallel inverters, except for DC solar inputs"*). PV strings are **not** shared: a dead inverter takes its own strings with it.

Fault codes (§8.1)

Code	Name	Meaning in a parallel system
F29	Parallel_Canbus_Fault	Lost parallel comms. *"Check cables, and Modbus addresses."* Normal for a moment during power-up until all units are on.
F41	Parallel_System_Stop_Fault	*"If one system faults in parallel, this normal fault will register on the other units as they disconnect from the grid."* The healthy units are reporting a neighbour's failure.
F46	Battery_Backup_Fault	*"Cannot communicate with other parallel systems. Verify that the Master is set to 1, the Slaves are set to 2–9, and the Ethernet cables are connected."*
F61	Button_Manual_OFF	*"The parallel Slave system turned off without turning off the Master."* Someone pressed a slave's power button.
F58	BMS_Communication Fault	Unit is in lithium BMS mode but has no BMS link — expect this on slaves if the master (which holds the CAN cable) goes down.

Ground rules

- **Power-cycle order (§2.12, parallel):** OFF — AC breakers (GRID, GEN, LOAD), then PV DC disconnect, then power button, **master first, then slaves**; battery breakers last, any order; wait ~1 min. ON — reverse it: **slaves first, master last**. *"Inverters will likely fault momentarily with F29 and F41 codes until all inverters are ON."* (§5.2 step 6)
- **DIP switches (§5.1):** termination on the two ends of the parallel daisy-chain. Two units: **both ON**. Three or more: first and last ON, middle units OFF. Any change to the chain means re-checking these. In a 3+ system, a dead middle unit must be cabled around (Parallel_1/Parallel_2 ports) and the two new ends re-terminated.


DIP Switch Configuration for Parallel Systems

In parallel systems, set the "DIP Switches" as shown, according to the table below.

Inv 1 (Master)	Inv 2	Inv 3	Inv 4	Inv 5	Inv 6	Inv 7	Inv 8	Inv 9	Inv 10	Inv 11	Inv 12
OFF											
ON	ON										
OFF	ON	OFF									
OFF	ON	ON	OFF								
OFF	ON	ON	ON	OFF							
OFF	ON	ON	ON	ON	OFF						
OFF	ON	ON	ON	ON	ON	OFF					
OFF	ON	ON	ON	ON	ON	ON	OFF				
OFF	ON	ON	ON	ON	ON	ON	ON	OFF			
OFF	ON	ON	ON	ON	ON	ON	ON	ON	OFF		
OFF	ON	ON	ON	ON	ON	ON	ON	ON	ON	OFF	
OFF	ON	ON	ON	ON	ON	ON	ON	ON	ON	ON	OFF

Parallel systems with 2 inverters must have their DIP switches in the ON position

Installation Manual Rev 13, §5.1. Two-inverter systems: both switches ON.

- **Firmware:** all units in parallel must show the same COMM and MCU version (§5.1 A). A replacement unit must be updated to match *before* it joins the chain.
- **Five strikes.** Self-clearing alarms auto-restart up to 5 times; the fifth stop on the same alarm locks the unit out until it is manually reset and the cause is cleared (installer's deck). A unit that is "dead" may just be locked out — read its alarm list before condemning it.

- **Batteries up before inverters.** Pytes V5 manual §5.4: power all packs, press the master pack's SW for 1 s, wait for every pack to show lights, *then* power the inverters — the inverter capacitors' inrush can shock an unready pack. Shut down in the opposite order (inverters first, then hold the master pack's SW 3 s).
- **Never run parallel mode without a battery bank** (§5.1) — irrelevant during a failure but the reason you don't "just" pull the battery to isolate a unit.
- **Charge current:** Gen/Grid "A" on the Battery Setup page is **per inverter** — the total into the bank drops automatically with one unit gone. Raise the per-unit value only if the generator can carry it.
- **Master-failure stopgap:** the generator is on the GRID input of every unit (§5.1 C), so surviving slaves are physically fed the moment it runs. Whether a slave's logic qualifies that input with no master on the chain is **not stated in the manual** — confirm with Sol-Ark support before it is needed.
- **SolarAssistant:** no reconfiguration after a failure or promotion. `/inverter/settings` shows the promoted unit as master; the dead unit's column stays stale. If low-SOC automations key off a specific inverter's values, re-point them.

Capacity

Parallel Systems Sol-Ark 15K-2P-LV @ 120/240 V Split-Phase

# of inverters in parallel	Continuous output power with PV (kW)	Continuous output power with batteries (kW)	Max. Grid Passthrough Current (A)	Peak power 10 sec (kVA)
1*	15	12	200	24
2	30	24	400	48
3	45	36	600	72
4	60	48	800	96
5	75	60	1000	120
6	90	72	1200	144
7	105	84	1400	168
8	120	96	1600	192
9	135	108	1800	216
10	150	120	2000	240
11	165	132	2200	264
12	180	144	2400	288

§5.1 table: per unit 15 kW continuous with PV, 12 kW on battery, 200 A grid passthrough, 24 kVA for 10 s.

Promotion checklist (technician)

1. Full power-down per §2.12. 2. Battery CAN → new master's Battery CANBus jack; two-wire start → pins 7 & 8; CT pair (grid-tied only). RS485 to SolarAssistant unchanged.
2. Parallel tab: Master, Modbus SN 01; other slaves stay 2…n. 4. Chain/termination as above. 5. Power up slaves first, master last. 6. Verify Grid type, Battery Setup (capacity, Shutdown/Restart, charge A, BMS Lithium Batt 00, Activate Battery), Gen Start %, Gen down time, Max Gen Runtime, TOU `Charge` ticks; confirm slaves report the same (§4 step E). 7. Gen Force → Gen Signal ≤2 min → start → clear. 8. Li-Batt Info live, no F58. Replacement unit: match firmware, Parallel ✓, vacated SN, DIP per position, cable in, full power cycle, verify settings copied. Never two units at SN 1.

Two-inverter quick card

	Slave (SN 2) dies	Master (SN 1) dies
Capacity left	15 kW PV / 12 kW battery	15 kW PV / 12 kW battery
Gen auto-start	still works	**lost** until promotion
BMS comms	still works	**lost** until CAN lead moved
Chain work	none; survivor's DIP stays ON	set SN 2 → 1; move CAN lead + gen start wires; DIP stays ON
Power-up order	just the survivor	just the survivor (it is now the master)
Expect	F41 logged on survivor	F41, F29/F46, F58 logged on survivor

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D. Solar panels — technician reference

Backs runbook page `cheatsheet-array-maintenance`.

Sources: installer's customer-training deck (as-delivered version); page 08 PV-forecast notes; tilt analysis on the site page.

Seasonal tilt. Two-switch schedule, PVGIS-derived: summer latitude −15° to −20° (not below ~15° on a bifacial ground mount, to keep rear-side gain and rain shedding), winter latitude +15° to +20° or the rack's maximum. For this site: ≈15° early April, ≈55° early October; ~+4.5 % front-side annual yield vs fixed 32°, Nov–Jan each up ~10–12 %. The SolarAssistant forecast has no seasonal model; `Configuration → Advanced → Solar PV → Tilt` must be updated on every move (Disconnect on the Devices panel → edit → Save → Connect). Re-torque rack hardware after each move; recheck before the first winter storm.

Array wiring. Strings of 7 in series along the top and bottom of each mount; top and bottom strings paralleled at the ends; one 600 V DC breaker per 14-panel group in a Midnite combiner box at the end of each array. Three series switch points: combiner breaker → exterior wall disconnect → inverter PV disconnect. All strings are equal, so on a clear day all MPPTs should read alike; 0 V = switch point open; low = shade, soiling or a failed module in that group. The single-line drawing is engraved on the exterior wall.

Panel replacement. Module: Trina TSM-NE09RC.05 410 W (Voc 50.1 V, 7 in series ≈ 350 V, up to ~390 V cold). Sinclair Designs mid- and end-clamp torque 20 ft-lb. MC4 connectors: Stäubli preferred; wire-nut joints are emergency-only.

Bifacial note. 65 ±10 % bifaciality; keep the ground under the array mowed and light-coloured; do not flatten below ~15° in summer.

Storm. F63 Arc_Fault after lightning is frequently spurious; clear manually (*Clear Arc Fault*) and watch for recurrence, which indicates a real connector fault.

E. Maintenance calendar — technician reference

Backs runbook page `cheatsheet-maintenance-calendar`.

Sources: Wildcat Patriot / Hyundai O&M manual Table 37 (general maintenance schedule) and Table 38 (log); Sol-Ark *15K Installation Manual* Rev. 13 §4 (weekly generator exercise, default Monday 08:00 for 20 min); installer's training deck; tilt analysis on the site page.

Table 37 mapping. Daily: walk-around, air-cleaner restriction indicator, oil level, coolant level, water-separator drain, hoses/clamps, radiator fins. First 50 h: fuel filters, belt adjust, oil + filter. Every 250 h / 1 yr (whichever earlier): fuel primary + secondary filters, belt adjust, oil + oil filter, centrifugal filter clean, CCV pipe drain (4K1080TA1 engine only). Every 500 h / 2 yr: air-cleaner element, belt replace, hoses/clamps, valve clearance, fasteners, mounts, starter, charging alternator, water pump, thermostat. 1500 h column: no items for these engines. The spec page's "oil change 500 h" is superseded by Table 37's 250 h / 1 yr column. Engine for the 40 kW set: HDI DM03PG, oil capacity 2.5 gal / 9.7 L, fuel 3.6 / 2.8 / 2.0 gal/h at 100 / 75 / 50 % load.

Tilt schedule. Two-switch: ≈15° early April, ≈55° early October (rack maximum if lower). SolarAssistant `Configuration → Advanced → Solar PV → Tilt` must follow every move (Disconnect → edit → Save → Connect). Expected ~+4.5 % annual front-side yield vs fixed 32°, concentrated Nov–Jan.

Generator cold-weather dependencies. NEMA 5-15 120 V inlet on the power-panel door feeds the magnetic sump heater (active near/below 32 °F, 1000–1500 W) and the built-in automatic starting-battery charger. External Battery Tender Plus 12 V / 1.25 A (Deltran 022-0185G-DL-WH) on the starting battery as a second maintainer. Anti-gel required; fuel-pump damage from gelled fuel is not warranted. Initial fill 115 gal, treated.

Warranty horizons. Pytes 10 yr; Sol-Ark 10 yr (includes lightning damage on the EMP-hardened units); panels 25 yr to ≥80 %; battery warranty excludes routine discharge below 10 %.